33kV INCOMING SUPPLY FROM F00-SW-010 PANEL B19
PLANT 33kV DISTRIBUTION SWITCHBOARD
2000-F00-SLD-EL-10005

F16-TR-011
33 / 11kV
10000kVA
Dyn11
z=8%
KNAN

F16-NR-011
FLOTATION 33/11kV
TRANSFORMER F16-TR-011
NEUTRAL EARTHING RESISTOR
2000-EL011-B02-10004

30/0.05A
EFCT

25A FOR 10s
254Ω +/-5%

SENSING
ER-15kV

F16-SW-011
FLOTATION 11kV SWITCHBOARD PANEL A09
2000-EL003-F16-10009

EARTH BAR

QCE
630A

BAD

PFV1

BCT1 600 / 1A 2.5VA CL 0.2
BCT2 600 / 1A 2.5VA CL 0.2
BCT3 600 / 1A 2.5VA CL 0.2
SEL 735

BCT1 600 / 1A 2.5VA 5P20
BCT2 600 / 1A 2.5VA 5P20
BCT3 600 / 1A 2.5VA 5P20
SEL 751

2000-EL003-F16-10009
SHEET 16

-QAB
1250A

11kV 1250A 3-E 25kA FOR 3s

BCF2
(PART OF)

F16TR011-JB
FLOTATION TRANSFORMER F16-TR-011 PROTECTION DEVICES TERMINAL BOX
2000-EL011-D02-10002

PRESSURE RELIEF VENT TRIP
OIL LEVEL ALARM (MINIMUM)
OIL LEVEL ALARM (MAXIMUM)
OIL TEMPERATURE ALARM (100°C)
OIL TEMPERATURE TRIP (110°C)
WINDING TEMPERATURE ALARM (110°C)
WINDING TEMPERATURE TRIP (125°C)
BUCHHOLZ ALARM
BUCHHOLZ TRIP

F16-SW-011
FLOTATION 11kV SWITCHBOARD PANEL A09
2000-EL003-F16-10009

BCF1 (PART OF)

1545 NOTE 4 — X2 10 — 48VDC CONTROL
1555 NOTE 4 — X2 55 — D10 IN409 TRANSFORMER PROTECTION TRIP NOTE 3
1605 NOTE 4 — X2 14 — 48VDC CONTROL
1615 NOTE 4 — X2 59 — D14 IN413 TRANSFORMER OIL LEVEL ALARM NOTE 3
1635 NOTE 4 — X2 15 — 48VDC CONTROL
1645 NOTE 4 — X2 60 — D15 IN414 TRANSFORMER OIL TEMPERATURE ALARM NOTE 3
1695 NOTE 4 — X2 16 — 48VDC CONTROL
1705 NOTE 4 — X2 61 — A10 IN101 TRANSFORMER WINDING TEMPERATURE ALARM NOTE 3
1765 NOTE 4 — X2 62 — A12 IN102 BUCHHOLZ ALARM NOTE 3

SHEET 13

F16SR001-LCS
SWITCHROOM EMERGENCY SHUTDOWN
PUSHBUTTON (LOCATED IN AIR LOCK)
2000-F16-TER-EL-10201

TO GENERATOR INCOMER 2000-F16-SCM-EL-10129

E-SHUT

1B35 NOTE 4 — X3 23
1B55 NOTE 4 — X3 33
SWITCHROOM EMERGENCY SHUTDOWN PUSHBUTTON CONNECTION TO THE VCB 1st TRIP CIRCUIT SHEET 8

1B75 NOTE 4 — X5 50
1B95 NOTE 4 — X5 51
NOTE 5

TO F00-SW-010 - PLANT 33kV HV DISTRIBUTION SWITCHBOARD FEEDER 2000-F00-SCM-EL-10137

1955 — X5 34
1965 — X5 35

BCF1 (PART OF)
C03 OUT302
C04
PROTECTION TRIP UPSTREAM

F16-RIO-100
11kV SWITCHROOM RIO PANEL
2000-F16-TER-IC-10014

X2 30 — F16SW011OP1+48VDC — R1 11 — 1A — 1B — 18/03/01
X2 63 — F16SW011OP1 — R1 14
DCS REMOTE CB OPEN

X2 31 — F16SW011CL1+48VDC — R2 11 — 2A — 2B — 18/03/02
X2 64 — F16SW011CL1 — R2 14
DCS REMOTE CB CLOSE

NOTES

1. WIRE NUMBER FERRULES ON TERMINAL STRIP SHALL BE PREFIXED WITH EQUIPMENT NUMBER e.g. F16SW011-1200, UNLESS OTHERWISE NOTED.
2. EACH WIRE SHALL BE IDENTIFIED AT BOTH ENDS WITH FERRULES.
3. ALL ALARMS AND TRIPS SHALL BE FAIL-SAFE. NOTE THAT THE PROTECTION RELAY INPUTS SHALL BE INVERTED, I.E. ALARM AND TRIP WHEN CONTACT IS OPENED.
4. NOTE THAT CROSS SECTIONAL AREA OF CABLES TO THE TRANSFORMER MARSHALLING PANEL FROM F00-SW-010 ARE INCREASED IN SIZE - REFER TO THE CABLE SCHEDULE.
5. E-SHUT CIRCUIT IS FAIL SAFE. NORMALLY CLOSED CONTACTS WILL BE OPEN WHEN THE E-SHUT BUTTON IS UN-ACTIVATED AND THE CIRCUIT HEALTHY.

LEGEND

- xx/xx/xx — DCS DIGITAL INPUT
- xx/xx/xx — DCS DIGITAL OUTPUT
- xx/xx/xx — DCS ANALOGUE INPUT
- xx/xx/xx — DCS ANALOGUE OUTPUT
- xx/xx/xx — DCS ANALOGUE RTD INPUT
- MCC TERMINAL
- VSD TERMINAL
- FIELD TERMINAL
- DCS OR I/O TERMINAL
- MARSHALLING TERMINAL
- (x) DEVICE TERMINAL

PANEL A09

2000-F16-SCM-EL-10117_Rev1

Rougher & scavenger flotation F16-SW-011 incomer 1 schematic diagram

O:\CAD PROJECTS\235 NORTHERN STAR RESOURCES LTD\235177 KCGM GROWTH PROJECT\03 DELIVERABLES\EL\F16 - ROUGHER & SCAVENGER FLOTATION\2000-F16-SCM-EL-10117.DWG 09SEP25 6:42AM HOTSEAT04